P1 is not a construction theorem

Fixed fold geometry, arbitrary sign-conforming weights: the class dies

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Abstract

Round 400 found that dead χ satisfy $\mathrm{P1} = \mathrm{ind}_-(R_{N-3} - \frac{1}{2}I) \le 1$ and that only the position-scramble breaks it. The natural suspicion (logged as DCCLXVII) is that P1 is a *construction* Satz of the fold mask, as the r391 white-block bounds were, so that the whole arithmetic load of the $R^\dagger$ route would sit in sch < 0. One lemma is tested: P1 holds for every sign-conforming reweighting of a frozen mask. It is refuted. Permute / Gauss / exponential / Dirichlet rescalings of the von Mangoldt weights at *fixed* positions give ind_- of order 10–25 on FRAME-A (median-flatten: $\mathrm{ind}_- = 2$). A relative 10^{-4} jitter already tips a disclosed draw. The separating invariant is the weight-to-position assignment, not the sign mask. No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Proposition 1 is a finite machine census that refutes the lemma. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 The lemma

Write $R = R_{N-3}$ for the dual Christoffel–Darboux Gram of the first $N_w - 3$ dual OP columns on the hole nodes Y, and $A_0 = R - \frac{1}{2}I$. P1 is $\mathrm{ind}_- A_0 \le 1$, equivalently $\lambda_2(R) \ge \frac{1}{2}$. The construction class is: freeze the cosine nodes and the sign pattern that splits them into μ-atoms and ν-holes, and vary the positive weights arbitrarily (permutations within each side, log-Gaussian, exponential, Dirichlet, median-flatten). This is the r391 protocol, pointed at A_0 rather than at the Fejér energy $B(\Sigma,\Sigma)/D$. It is *not* the r400 scramble, which redraws *positions*.

Lemma 1 (rank-one interlacing)**.** *A symmetric rank-one update changes* ind_- *by at most one. Over* $\mathbb{Q}$*:* $\mathrm{diag}(-1,-1) + e_0 e_0^T = \mathrm{diag}(0,-1)$.

Lemma 2 (PSD need not finish a lift)**.** *A positive-semidefinite update of a negative-definite matrix may leave every negative eigenvalue negative. Over* $\mathbb{R}$*:* $\mathrm{diag}(-1,-1) + 0.1\,I$ *still has* $\mathrm{ind}_- = 2$.

Lemma 3 (scale invariance of R)**.** *Joint rescaling* $(w_\mu, w_\nu) \mapsto c(w_\mu, w_\nu)$ *leaves* R *unchanged. On FRAME-A,* $\|R - R(7w)\|_F = 3.0 \cdot 10^{-13}$.

The dual weights of $|cw|$ scale so that the CD Gram on Y is homogeneous of degree zero. Relative μ/ν scale is *not* free: $\mu \mapsto 3\mu$ sends FRAME-A to the vacuous chart ($\mathrm{ind}_- = 0$); $\nu \mapsto 3\nu$ yields $\mathrm{ind}_- = 65$.

2 The class, refuted

Proposition 1 (construction class). *P1 does not hold on the construction class. On FRAME-A, four seeds of a mask-preserving permutation of the von Mangoldt weights give* $\mathrm{ind}_- \in \{20, 21, 21, 22\}$*; log-Gaussian* $\{18, 19, 22, 24\}$*; exponential* $\{11, 14, 15, 11\}$*; Dirichlet* $\{11, 12, 18, 12\}$*; median-flatten (deterministic)* $\mathrm{ind}_- = 2$*. A disclosed six-draw at relative amplitude* 10^{-4} *already has one sample with* $\mathrm{ind}_- > 1$*. On the sample* $\{9, 22, 33, 78, 82\}$ *MAIN stays at* $\mathrm{ind}_- = 1$ *and a single permutation yields* $24, 30, 44, 96, 102$*.*

The sign mask is therefore not the invariant. Neither is positivity, nor block cardinality, nor joint scale (Lemma 3). What separates the P1 worlds from the scramble is the *assignment* of von Mangoldt (or χ) masses to the frozen nodes. Dead χ_3-15 has $\mathrm{ind}_- = 1$ on its true weights and $\mathrm{ind}_- = 20$ after a permutation of those weights: it sits in P1 for the same arithmetic reason MAIN does, not because it shares a construction class with MAIN.

3 Saturation and the Gram candidate

The Bernstein–Szegő dual on Y (r390) has $\mathrm{ind}_-(R^{\mathrm{ref}} - \frac{1}{2}I) = 49$ at FRAME-A and 13 eigenvalues of R^{ref} at $\frac{1}{2}$ to 10^{-10}. Those 13 modes are not a single exact half-filling that the source lifts infinitesimally: the count grows ($13, 27, 38, 62, 65$ on the sample). The identity "$48 = 49 - 1$" is the tautology killed $= \mathrm{nneg}_{\mathrm{ref}} - \mathrm{ind}_- A_0$ on MAIN, not a class law — $\mathrm{nneg}_{\mathrm{ref}}$ itself is $49, 48, 54, 124, 141$ on the same sample, and a permutation leaves tens of negatives.

Write $B = R - R^{\mathrm{ref}}$ and $M = P_-^{\mathrm{ref}} B P_-^{\mathrm{ref}}$. On MAIN, M is positive definite ($\mathrm{ind}_- M = 0$, 49 positive eigenvalues, residual 0, spectrum in $[0.023, 0.710]$): the restriction is a Gram. Omitting that Gram part returns $\mathrm{ind}_- = 49$. Keeping *only* that Gram part leaves $\mathrm{ind}_- = 13$ (Lemma 2 in situ: the cross terms of the full B finish the lift down to 1). Off the MAIN weights the Gram property dies with P1 (permutation residual 0.14, $\mathrm{ind}_- M = 6$; scramble residual 0.15, $\mathrm{ind}_- M = 8$). So the Gram observation is a MAIN-weight census, not a construction Satz and not a proof of P1.

4 Named kills

Position-scramble (r400): $\mathrm{ind}_- = 21$. It breaks P1, but so does a permutation that *keeps* the mask — scramble is not the unique kill and is not the separating invariant. Two-period $S = 21$: $\mathrm{ind}_- = 4$, killed 0. Sign-flip of the union weights redraws Y and yields $\mathrm{ind}_- \sim 90$. Omit-Gram mutant: $\mathrm{ind}_- = 49$, as required. Dead χ stay at $\mathrm{ind}_- \leq 1$ on their true weights and leave the class under the same permutation.

Machine check

Standalone: rank-one interlacing, PSD-incomplete lift, threshold $\frac{1}{2}$. Construction: FRAME-A inertia and Gram, scale, μ/ν split, five-law weight-rand, mild jitter, scramble, two-period, dead-χ weights versus permutation. Discovery census: `p1_construction_probe.py` (`--smoke` / full), sample of five rungs plus EXT $k_z = 82$. Exit line: `P1 CONSTRUCTION VERIFIED`.

Claim boundary

Finite identities for interlacing, an incomplete PSD lift, and scale invariance of R; a named census that P1 fails as soon as the von Mangoldt profile is disturbed on a frozen mask. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

[1] S. Hamann, *G^{μ}_{ε} / Bernstein–Szegő Fejér*, sealed probe, August 2026.

[2] S. Hamann, *Construction-pure* (R) *and* (L), standalone note, August 2026, `rh/problem/construction_rl.tex`.

[3] S. Hamann, *Bulk one-defect*, standalone note, August 2026, `rh/problem/bulk_one_defect.tex`.

[4] S. Hamann, *The augmented edge signature*, standalone note, August 2026, `rh/problem/edge_signature.tex`.